DEEP LEARNING LAB

Algorithms for Selected Experiments

Experiment 1 — Regression

Algorithm

Step 1: Start.

Step 2: Import the required libraries and load the Advertising dataset.

Step 3: Select TV advertising as the input feature and Sales as the target for Simple Linear Regression.

Step 4: Split the data into training and testing sets.

Step 5: Train a Linear Regression model, predict Sales, and calculate MSE and R^2 score. Plot the data and regression line.

Step 6: Select TV, Radio, and Newspaper as input features for Multiple Linear Regression.

Step 7: Train the model, predict Sales, calculate MSE and R^2 score, and plot actual versus predicted Sales.

Step 8: Transform the input feature into polynomial features of degree 2.

Step 9: Train the Polynomial Regression model, predict Sales, calculate MSE and R^2 score, and plot the polynomial curve.

Step 10: Display the results and stop.

Experiment 2 — K-Nearest Neighbors (KNN) Classification

Algorithm

Step 1: Start.

Step 2: Import the required libraries and generate a two-class classification dataset.

Step 3: Split the dataset into training and testing sets.

Step 4: Create a KNN classifier with K = 5.

Step 5: Train the classifier using the training data.

Step 6: Predict the class labels of the test data.

Step 7: Calculate and display accuracy, precision, and recall.

Step 8: Stop.

Experiment 3 — Naive Bayes Classification (NBC)

Algorithm

Step 1: Start.

Step 2: Import the required libraries and load the Iris dataset.

Step 3: Separate the features and target labels.

Step 4: Split the data into training and testing sets using stratified sampling.

Step 5: Create and train a Gaussian Naive Bayes classifier.

Step 6: Predict the classes of the test samples.

Step 7: Calculate and display accuracy, weighted precision, and weighted recall.

Step 8: Stop.

Experiment 5 — Hierarchical Agglomerative Clustering (HAC)

Algorithm

Step 1: Start.

Step 2: Import the required libraries and generate 150 sample points with three cluster centers.

Step 3: Compute the hierarchical linkage matrix using Ward linkage.

Step 4: Plot the dendrogram to show the order in which clusters are merged.

Step 5: Create an Agglomerative Clustering model with three clusters and Ward linkage.

Step 6: Fit the model and obtain a cluster label for each data point.

Step 7: Plot the data points using different colors for the assigned clusters.

Step 8: Display the dendrogram and clustering plot, then stop.

Experiment 6 — K-Means Clustering

Algorithm

Step 1: Start.

Step 2: Import the required libraries and generate 300 sample points with four cluster centers.

Step 3: Create a K-Means model with K = 4, K-Means++ initialization, and 10 initializations.

Step 4: Fit the model and assign each data point to a cluster.

Step 5: Obtain the cluster centroids and calculate the inertia (within-cluster sum of squares).

Step 6: Display K and the inertia value.

Step 7: Plot the clusters in different colors and mark the centroids.

Step 8: Display the plot and stop.